1) $\int_2^3 f\,dx = \int_2^6 f\,dx + \int_6^0 f\,dx + \int_0^3 f\,dx$

$= \int_2^6 f\,dx - \int_0^6 f\,dx + \int_0^3 f\,dx$

$= -2 - 8 + 4 = -6.$

2)

a) $\frac{d}{dx}\int_{x^2+3x}^{x^3} \ln u\,du = \ln(x^3)\frac{d}{dx}(x^3) - \ln(x^2+3x)\frac{d}{dx}(x^2+3x)$

$= 3x^2\ln(x^3) - (2x+3)\ln(x^2+3x)$

b) $\frac{d}{dx}\int_{2x}^{3} \frac{\sec^2 t}{\ln t}\,dt = -\frac{\sec^2(2x)}{\ln(2x)}\cdot\frac{d}{dx}(2x) = \frac{-2\sec^2(2x)}{\ln(2x)}$

c) $\frac{d}{dx}\int_2^5 e^{x^2}\,dx = \frac{d}{dx}(\text{constante}) = 0$

d) $\frac{d}{dx}\int_2^x \frac{5}{\sqrt{1-t^2}}\,dt = \frac{5}{\sqrt{1-x^2}}$

3) a) $\int_1^e x\ln x\,dx$ integrando por partes $f' = x$, $g = \ln x$ $\Rightarrow$ $f = \frac{x^2}{2}$, $g' = \frac{1}{x}$

daí $\int_1^e x\ln x\,dx = \frac{x^2}{2}\ln x\Big|_1^e - \int_1^e \frac{1}{2}\cancel{x^2}^{\,x}\cdot\frac{1}{\cancel{x}_{\,1}}\,dx$

$$= \frac{e^2}{2}\ln e - \cancel{\frac{1}{2}\ln 1}^{\,0} - \frac{1}{2}\int_1^e x\,dx$$

$$= \frac{e^2}{2} - \frac{1}{2}\left(\frac{x^2}{2}\Big|_1^e\right) = \frac{e^2}{2} - \frac{1}{2}\left(\frac{e^2}{2} - \frac{1}{2}\right) = \frac{e^2}{4} + \frac{1}{4} = \frac{e^2+1}{4}$$

b) $\int \frac{x^2-1}{x^2+1}\,dx = \int \frac{x^2+1-2}{x^2+1}\,dx = \int \cancel{\frac{x^2+1}{x^2+1}} - \frac{2}{x^2+1}\,dx$

$$= \int 1 - \frac{2}{x^2+1}\,dx = \int 1\,dx - 2\int \frac{1}{x^2+1}\,dx = x - 2\arctan x + c$$

c) $\int \operatorname{sen}^3 x\,dx = \int \operatorname{sen}^2 x \operatorname{sen} x\,dx = \int (1-\cos^2 x)\operatorname{sen} x\,dx$

$$= \int (\cos^2 x - 1)(-\operatorname{sen} x)\,dx$$

fazer $u = \cos x \Rightarrow du = -\operatorname{sen} x\,dx$ logo

$$\int (\cos^2 x - 1)(-\operatorname{sen} x)\,dx = \int u^2 - 1\,du = \frac{u^3}{3} - u + c$$

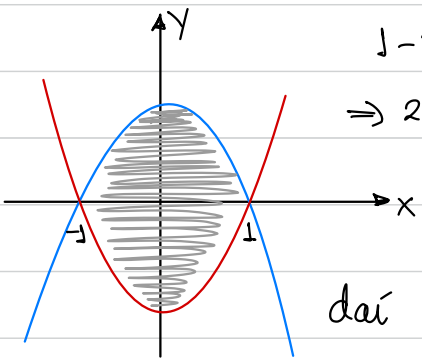
$$= \frac{\cos^3 x}{3} - \cos x + c.$$

d) $\displaystyle\int \frac{2x}{\sqrt{1-x^2}}\,dx$ fazer $u = 1-x^2 \Rightarrow du = -2x\,dx$

daí $\displaystyle\int \frac{2x}{\sqrt{1-x^2}}\,dx = -\int \frac{-2x}{\sqrt{1-x^2}}\,dx = -\int \frac{du}{\sqrt{u}} = -\int u^{-1/2}\,du$

$$= -\frac{u^{1/2}}{1/2} + c = -2\sqrt{u} + c = -2\sqrt{1-x^2} + c$$

4)



$1 - x^2 = x^2 - 1 \Rightarrow 2x^2 - 2 = 0 \Rightarrow 2(x^2-1) = 0$

$\Rightarrow 2(x-1)(x+1) = 0 \Rightarrow x = -1$ ou $x = 1$

daí $\displaystyle A = \int_{-1}^{1} (1-x^2) - (x^2-1)\,dx = \int_{-1}^{1} -2x^2 + 2\,dx$

$$= 2\int_0^1 -2x^2 + 2\,dx = 2\left(-\frac{2x^3}{3} + 2x \Big|_0^1\right)$$

$$= 2\left(-2/3 + 2\right) = 2\left(4/3\right) = 8/3$$